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Lesson plan

Name of the Programme:	**B.Tech (Hons.)**	
Title of the Course:	**Cloud Computing and Big Data**	
Course code	**CS2500**	
Semester:	**III**	
Names of the Course Instructor(s)	**Dr. Ananya Chakraborty**	
Course credit	**No. of hours per week (2+0+2)**	**Total no. of Teaching hours**
03	**04 Hours**	**60 Hours**

Session	Module No.	Topic	Pedagogy /activities	Pre-reading/reference
1.	1	Introduction to Computing, Cloud, Grid, Edge, Fog, Utility	PPT	Buyya – Ch. 1, Bahga – Ch. 1
2.		Characteristics of cloud computing; Types of Cloud Computing Services: PaaS, IaaS, and SaaS,	PPT and discussion	Buyya – Ch. 1, Bahga – Ch. 2
3.		Lab 1: Exploring CloudSim: Understanding Cloud Simulation Through a Basic Setup	Guided Installation + Simulation Demo	Review of fundamental cloud computing concepts
4.				
5.		Types of Cloud Deployment Models: Public Clouds, Private Clouds, Community Clouds, and Hybrid Clouds	PPT and discussion	Buyya – Ch. 1
6.		Multi-tenancy	PPT	Buyya Ch. 1
7.		Lab 2: Understand CloudSim configurations and scheduling algorithms.	Code walkthrough of default scheduling followed by guided implementation and testing of a custom algorithm.	CloudSim class structure and basic scheduling algorithms
8.				
9.	2	Introduction to virtualization technologies, Benefits of Virtualization, Virtual Machines	PPT	Buyya – Ch. 3, Bahga – Ch. 2
10.		Type 1 and Type 2 Hypervisors.	PPT	Buyya – Ch. 3

11.		Lab 3: Installing a Type 2 Hypervisor	Guided Installation+ Demo	Concept of virtualization
12.				
13.		Taxonomy of virtualization techniques: machine reference model.	PPT	Buyya – Ch. 3
14.		Hardware virtualization techniques (full, para, hardware-assisted)	PPT	Buyya – Ch. 3, Velte- Ch. 1
15.		Lab 4: Create and Manage Virtual Machines	Hands On+ Demo	Basic operations in a Type 2 hypervisor interface
16.				
17.		OS level virtualization, Programming level virtualization, Application virtualization, storage virtualization.	PPT	Buyya – Ch. 3
18.		Network virtualization (SDN, NFV), Desktop virtualization, Application server virtualization.	PPT	Buyya – Ch. 6
19.		Lab 5: Install Docker and Run Basic Containers	Guided Installation + Hands-on Commands	Introduction to containers and Docker architecture.
20.				
21.		Containerization versus virtualization, Introduction to docker	PPT	Buyya – Ch. 5
22.		Docker hub and Kubernetes. Container orchestration concepts.	PPT	-
23.		Lab 6: Build and Run Custom Docker Images	Dockerfile Writing + Hands-on Build & Run	Docker commands
24.				
25.	**3**	Cloud Reference Architecture	PPT	Buyya- Ch. 1
26.		IAM (Amazon IAM)	PPT	Bahga – Ch. 3
27.		Lab 7: Launch an AWS EC2 Instance and Connect via SSH. Understanding AWS IAM Users, Groups, and Roles	Demo	Overview of AWS EC2
28.				
29.		Compute services (AWS EC2, AWS Lambda)	PPT	Bahga – Ch. 3, Velte- Ch. 3
30.		Storage Services (AWS S3)	PPT	Bahga – Ch. 3, Velte- Ch. 3
31.		Lab 8: Use AWS S3 to Store and Retrieve Data	Guided S3 Setup + Hands-on Data Operations	Basics of AWS S3
32.				
33.		Database services (Amazon RDs, Dynamo)	PPT	Bahga- Ch. 3

34.		Content Delivery Services (Amazon CloudFront)	PPT	Bahga- Ch. 3
35.		Lab 9: Integrating AWS Lambda with Amazon DynamoDB	Guided Setup + Hands-on Querying	Basic concepts of AWS RDS
36.				
37.		Analytics Services (Hadoop Framework and AWS MapReduce),	PPT	Bahga- Ch. 3
38.		Deployment Services (Amazon Elastic Beanstalk, cloud formation).	PPT	Bahga- Ch. 3
39.		Lab 10: Building a Serverless Chatbot with API Gateway, Lambda, and Amazon Bedrock.	Guided Deployment + Environment Exploration	Overview of AWS Elastic Beanstalk
40.				
41.		Big Data, Concepts, Properties, characteristics, Applications and value of big data	PPT	Acharya – Ch. 1, 2
42.		Big data analytics lifecycle, Scalability of Big Data Analytics (Horizontal and Vertical)	PPT	Acharya – Ch. 3, 4
43.		Lab 11: MongoDB Installation and CRUD Operations	Guided Installation	Introduction to NoSQL
44.	**4**			
45.		Designing Data Analytics Architecture, Overview of NoSQL Database	PPT	Acharya – Ch. 3, 4
46.		NoSQL to Manage Big Data, Application of MongoDB in Big Data.	PPT	Acharya – Ch. 6
47.		Lab 11: MongoDB Installation and CRUD Operations (continued)	Hands-on CRUD Practice	Basic MongoDB concepts
48.				
49.		Introduction	PPT	Acharya – Ch. 5
50.		Hadoop and its Ecosystem	PPT	Acharya – Ch. 5
51.		Lab 12: Hadoop Setup and WordCount Program Execution	Hadoop Setup	Hadoop architecture review
52.				
53.		Hadoop Distributed File System	PPT	Acharya – Ch. 5
54.	**5**	MapReduce Framework and Programming Model	PPT	Acharya – Ch. 8
55.		Lab 12: Hadoop Setup and WordCount Program Execution (continued)	WordCount Demo	-
56.				
57.		Hadoop Yarn, Hadoop Ecosystem Tools	PPT	Acharya – Ch. 9
58.		HDFS Design Features, HDFS User Commands.	PPT	Acharya – Ch. 9

59.		Lab 12: Hadoop Setup and WordCount Program Execution (continued)	Demo	-
60.				

Assessment and Evaluation

Continuous Internal Evaluation (CIE) :70%	
Components of CIE	Weightage of each component
CIE-1 - Quiz	20 marks out of 70
CIE-2 - Theory	25 marks out of 70
CIE-3 - Lab execution & viva - Case Study	10 marks out of 70 15 marks out of 70

Semester End Exams (SEE): 30%	
Mode of Exam	Theory
Weightage of exam	30%

Course Outcomes- Progamme Outcomes Matrix

Program Outcome Course Outcome		PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO10	PO11	PO12
CO 1	Predict suitable cloud computing services and deployment models for different applications.	3	2	2	2	1							
CO 2	Integrate virtualization and containerization technologies	3	3	2	3	1				1			

	to implement CI/CD best practices for deploying cloud-based applications.												
CO 3	Distinguish the services offered by various Cloud Service Providers across Compute, Storage, Network, and Database categories.	2	1	2	3	1	1	1					
CO 4	Analyze the applications of big data analytics, and NoSQL databases to derive insights for decision-making.	3	3	2	3	1	2	1		1			
CO 5	Design solutions for processing big data using the Hadoop Framework in real-world applications.	3	3	2	1	1	1	1	1	1			2

Rubrics for evaluation of CIE (separate for each CIE component)

Total Marks		**20**
CIE-1 Components		**Rubrics for Assessment**
CIE-1	Quiz	Quiz will be conducted for 20 marks. There will be 10 1-mark questions and rest 10-marks theory questions. No negative marking for wrong answers.

Total Marks		25
CIE-2 Components		**Rubrics for Assessment**
CIE-2	Theory	Test will be conducted for 25 marks and will be evaluated according to Scheme of Evaluation prepared by course faculty.

Total Marks		25
CIE-3 Components		**Rubrics for Assessment**
CIE-3	Practical Component	Practical execution+ Viva = 10 marks ***(Refer Table-1)***
	Case Study Component	Course Case Study = 15 marks ***(Refer Table-2)***

Table-1

Practical Execution rubrics (Max: 6 marks)					
Sl. No	Criteria	Measuring Methods	Excellent	Good	Poor
1.	Understandin g of problem statements. (2 Marks)	Observations	Demonstrates clear understandin g of cloud/big data requirements and selects suitable approach (2 Marks)	Demonstrates basic understanding of the requirements; selects an approach that is appropriate but not optimal. (1 Mark)	Demonstrates little or no understanding of requirements; selects an incorrect or unsuitable approach. (.5 Marks)
2.	Execution (4 Marks)	Observations	Demonstrates optimized and efficient execution using cloud/big data tools (4 Marks)	Executes the task correctly with minor inefficiencies or minor errors; overall use of tools is adequate. (2 Marks)	Execution is incorrect, inefficient, or incomplete; improper or ineffective use of cloud/big data tools. (1 Marks)

Viva Voce Rubrics (Max: 4 marks)					
Sl. No	Criteria	Measuring Methods	Excellent (2 Marks)	Good (1 Mark)	Poor (0.5 Marks)
1.	Conceptual Understanding of Cloud/Big Data Techniques (2 Marks)	Viva Voce	Thorough explanation of concepts and tools used (2 Marks)	Adequate explanation with some gaps (1 Mark)	Unable to explain tools/concepts clearly (0.5 Marks)
2.	Application of Problem-Solving Techniques (2 Marks)	Viva Voce	Describes and justifies the selection of specific services (2 Marks)	Identifies applicable tools or methods with minor gaps in justification (1 Mark)	Lacks clarity or incorrect application (0.5 Marks)

Table-2

Case Study (Total: 15 Marks)					
Sl. No	Criteria	Excellent	Very Good	Good	Needs Improvement
1.	Report Quality & Documentation (4 Marks)	Exceptionally clear, well-structured, professionally formatted; covers all aspects with depth and originality. (4 M)	Well-written and logically structured with minor formatting or content issues. (3 M)	Adequate coverage, lacks depth or organization in parts. (2 M)	Incomplete, disorganized, or unclear documentation. (0–1 M)
2.	Conceptual Understanding (4 Marks)	Demonstrates thorough and critical understanding of key concepts, context, and issues in the case. (4 M)	Shows good grasp of the core ideas with a few minor gaps. (3 M)	Basic understanding, but with some misinterpretations. (2 M)	Misunderstands or misses core concepts or context. (0–1 M)
3.	Presentation (4 Marks)	Clear, engaging, and confident delivery; excellent use of visuals and time	Organized and clear presentation; minor issues with	Understandable but lacks engagement or has significant timing/design issues. (2 M)	Poorly delivered; unclear, unstructured, or heavily reliant on

		management; highly interactive. (4 M)	engagement or visuals. (3 M)		reading. (0–1 M)
4.	Viva (3 Marks)	Confident, clear, and engaging delivery; handles questions with insight and clarity. (3 M)	Well-organized presentation and competent response to questions. (2 M)	Some hesitation or unclear responses; moderate engagement. (1 M)	Poor delivery or unable to respond to questions effectively. (0 M)

Signature of the Course Faculty **Signature of the Program Director**